NCAS STUDENT CHALLENGE

Career Portfolio

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Electrical Engineering & Computer Science · Berkeley City College · NASA Community College Aerospace Scholar

Portfolio website: williamlebris.github.io/website/#/career-portfolio

1. Executive Summary

I am an Electrical Engineering and Computer Science student at Berkeley City College and Laney College with a 4.0 GPA, planning to transfer to a four-year university in 2027. My goal is to become an electrical and AI systems engineer who builds the hardware and software that turn spacecraft data into science for NASA missions.

My coursework in calculus, differential equations, linear algebra, physics with calculus, electrical engineering, assembly language, and data structures gives me that foundation. In summer 2026 I applied it as an AI Research Intern at the UC Berkeley Space Sciences Laboratory, which built the instruments for and operates NASA's Carruthers Geocorona Observatory.[1] I built computer vision pipelines that detect stars in the spacecraft's images to check its pointing and find optical distortions. This fall I am the Electrical Engineer on a team in L'SPACE Mission Concept Academy, a NASA-funded program[2] where students design a mission concept that follows the NASA mission life cycle.[3] As a NASA Community College Aerospace Scholar, I am connecting my coursework to NASA career pathways.[4] I also founded StudentLife OS, an AI platform launched in August 2025 that connects students with internships, scholarships, and research programs.[5]

These experiences prepare me for NASA roles in spacecraft power and avionics, flight and ground software, and science data analysis. My next steps are a NASA internship through the Office of STEM Engagement or the Pathways Program[6,7] and a bachelor's degree in electrical engineering and computer science.

2. Technical Skills

1. Python & Scientific Data Analysis

Developed through: SSL research internship; Data Structures & Algorithms; Object-Oriented Programming

NASA connection: Used to process and validate image data from NASA's Carruthers mission at UC Berkeley SSL. [1]

2. Computer Vision & Image Processing (OpenCV)

Developed through: Star detection and tracking at SSL; Healthy Plan AI project

NASA connection: Star trackers find a spacecraft's orientation by comparing a camera image to an onboard star catalog, the same idea behind my Carruthers pointing work. [8]

3. AI & Machine Learning (PyTorch, TensorFlow, scikit-learn)

Developed through: StudentLife OS AI application assistant; personal ML projects

NASA connection: NASA uses AI across its missions, from autonomous rover driving on Mars to satellite image analysis. [9]

4. Circuit Analysis & Design

Developed through: Intro to Electrical Engineering; General Physics with Calculus; L'SPACE spacecraft design

NASA connection: Every spacecraft subsystem relies on circuits that distribute power and carry commands and data. [10,11]

5. Spacecraft Electrical Power Systems

Developed through: Electrical Engineer role in L'SPACE Mission Concept Academy

NASA connection: Power generation, energy storage, and power management and distribution make up a spacecraft's power subsystem. [10]

6. Embedded Systems & Low-Level Programming (C/C++, Assembly, Arduino, Raspberry Pi)

Developed through: Assembly Language course; hardware and network repair internship at American Tech

NASA connection: Spacecraft avionics run flight software on embedded processors. [11]

7. Systems Engineering & the NASA Mission Life Cycle

Developed through: L'SPACE Mission Concept Academy training and team design work

NASA connection: NASA's Systems Engineering Handbook defines the life cycle phases and design reviews used on NASA flight projects. [12]

8. Engineering Mathematics & Modeling

Developed through: Calculus; Differential Equations; Linear Algebra

NASA connection: Linear algebra drives the camera geometry I used to measure pointing and distortion, and differential equations describe orbits and circuits.

9. Full-Stack Software Engineering (Next.js, React, TypeScript, PostgreSQL, Prisma)

Developed through: Building and launching StudentLife OS [5]

NASA connection: The same skills build the dashboards, databases, and data archives that mission teams and scientists rely on.

10. Cybersecurity & Secure Data Handling

Developed through: StudentLife OS security program; Mineral cybersecurity certificates

NASA connection: Federal agencies, including NASA, manage cyber risk with NIST standards such as the Cybersecurity Framework 2.0. [13]

3. Experiences & Projects



AI Research Intern

UC Berkeley Space Sciences Laboratory (SSL)

📍 Berkeley, CA 📅 Jun 2026 - Aug 2026

Experience: I worked at the UC Berkeley Space Sciences Laboratory, which built the instruments for and operates NASA's Carruthers Geocorona Observatory, a spacecraft that images Earth's outermost atmosphere from about a million miles away. [1,14] I built Python and OpenCV pipelines that detect and track the stars that appear in the spacecraft's ultraviolet camera images. By comparing where each star appears with where it should be, the pipeline measures where the spacecraft is pointing and reveals optical distortions in the cameras. I used these results to validate mission observations and improve the team's science data processing, and I earned SSL's certificate for Advancing Space Sciences through Undergraduate Research Experience.

NASA & industry alignment: Carruthers is a NASA heliophysics mission that studies how Earth's exosphere responds to space weather, knowledge that helps protect satellites and

astronauts traveling to the Moon under Artemis.[1,14] Accurate pointing and distortion correction turn raw images into reliable science, so this work supports the mission's core goal. The method uses the same star-matching idea as spacecraft star trackers,[8] and it shows how AI and computer vision can speed up analysis of large NASA datasets.[9] It prepared me for NASA roles in science data analysis, instrument calibration, and mission operations.



Electrical Engineer

NASA L'SPACE Mission Concept Academy

Virtual Sep 2026 - Present

Experience: L'SPACE (Lucy Student Pipeline Accelerator and Competency Enabler) is a NASA-funded, 15-week academy where interdisciplinary student teams develop a preliminary mission concept that follows the NASA mission life cycle.[2,3] As my team's Electrical Engineer, I am responsible for the spacecraft's electrical design: power generation and storage, the power budget, and how the avionics and science instruments connect. I work with teammates in science, mechanical, and systems roles to compare design options and keep the concept within mass, power, and cost limits. We document our design in technical reports that follow NASA review procedures, which strengthens my technical writing.[3]

NASA & industry alignment: NASA funds the academy through its Lucy mission to prepare students for careers on NASA missions.[2] Working through the mission life cycle gives me direct practice with the systems engineering process NASA uses on its flight projects.[12] Designing a spacecraft's power and avionics matches the work of NASA electrical and avionics engineers.[10,11] It also turns my Intro to Electrical Engineering and physics coursework into real design decisions.



Founder & Lead Engineer

StudentLife OS

San Francisco Bay Area, CA Launched Aug 2025

Experience: I founded and built StudentLife OS, a platform launched in August 2025 that connects students with internships, scholarships, research programs, and campus resources, and gives schools and employers tools to support them.[5] I designed the full stack (Next.js, React, TypeScript, PostgreSQL, Prisma) and an AI application assistant that

drafts answers only from a student's own saved data, so it never invents facts. I also built the platform's security program, including encryption, audit logging, and a self-assessment against the NIST Cybersecurity Framework 2.0.[13] I lead a team of student interns who help build and test new features.

NASA & industry alignment: NASA and the tech industry both need engineers who can ship secure, reliable software, and the same skills build mission dashboards and science data archives. NASA stresses using AI tools safely to support missions and research,[9] the same principle behind keeping students in control of every AI-drafted answer. Running a security program against NIST standards mirrors how federal agencies manage cyber risk.[13] The platform also helps community college students find STEM opportunities like NCAS, which supports NASA's goal of reaching more of these students.[4]

4. NASA Career Pathway Alignment

This table shows how my community college coursework builds the skills I have applied in real projects, and the NASA and aerospace roles those skills lead to.

Coursework	Skills built	Where I applied them	NASA / aerospace pathway
Intro to Electrical Engineering; General Physics with Calculus	Circuit analysis, power systems	L'SPACE Electrical Engineer	**Electrical / Avionics Engineer: spacecraft power and avionics**[10,11]
Data Structures & Algorithms; Object-Oriented Programming	Python, computer vision, AI, software design	SSL Carruthers internship; StudentLife OS	**Data Scientist / AI Engineer for science missions**[9]
Linear Algebra; Differential Equations; Calculus	Image geometry, mathematical modeling	Carruthers pointing and distortion analysis	**Guidance, Navigation & Control / Instrument Calibration**[8]
Assembly Language	Low-level programming, embedded systems	Hardware and network repair internship (American Tech)	**Flight Software / Embedded Systems Engineer**[11]
Engineering coursework + L'SPACE training	Requirements, trade studies, design reviews	L'SPACE Mission Concept Academy	**Systems Engineer**[12]

My path to NASA

> **2024 - 2027:** Berkeley City College & Laney College: Electrical Engineering & Computer Science coursework, 4.0 GPA
>
> **2026:** AI Research Intern at UC Berkeley SSL (NASA Carruthers); L'SPACE Electrical Engineer; NCAS Scholar; Stanford Small Group Scholars Program (data labeling for AI)
>
> **2027:** Transfer to a four-year Electrical Engineering & Computer Science program
>
> **2027 - 2029:** NASA Office of STEM Engagement internships and the NASA Pathways Program[6,7]
>
> **Career:** Electrical and AI systems engineer on NASA science missions, at a NASA center, a university lab like SSL, or an aerospace company

5. Resume

William Lebris Tissi Nguemetcha

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EDUCATION

Berkeley City College - Berkeley, CA — Expected Graduation Spring 2027

Electrical Engineering & Computer Science Major, GPA 4.0

Relevant Courses:

- Differential Equations, Calculus, Assembly Language, Intro to Electrical Engineering, Linear Algebra, General Physics with Calculus, Economics, Data Structures & Algorithms, Object-Oriented Programming.

SKILLS

Technical Skills:

- Advanced - C++, Java, Python, MySQL, HTML5, JavaScript, CSS, React, TypeScript, Node.js, PostgreSQL, Prisma, Sentry, Data Structures and Algorithms, Computer repairs, GitHub, Bluehost, Web Security, Circuit design, Hardware, OpenCV(image processing, feature detection), Solving Calculus problems, Microsoft Office, Git, GitHub, Adobe.

Soft Skills: Communication, Teamwork, Problem-solving, Adaptability, Leadership, Time Management, Critical thinking, Quick learner, Collaboration, Report writing and presenting, and Decisiveness, French & English Speaker, writer, Translator

RELEVANT ACADEMIC EXPERIENCE

Computer and Data Science Club - Vice President, — November 2024 - Present

- Acquired fundamental Python programming skills, including loops, functions, data types, and data structures

Math Engineering Science Achievement - Member — October 2024 - Present

- Collaborating on STEM projects, receiving academic support, and engaging in industry visits to companies like Meta.

WORK EXPERIENCE

NASA Community College Aerospace Scholars (NCAS) — October 2026 - Present

NASA

- Selected to participate in NASA's NCAS program, gaining exposure to NASA missions and aerospace engineering through hands-on engineering activities, collaborative mission design, and interaction with NASA professionals.

Electrical Engineer | NASA L'SPACE Academy — September 2026 - Present

NASA L'SPACE

- Selected to participate in NASA's L'SPACE Academy as an Electrical Engineer, collaborating on a simulated NASA mission project to develop engineering solutions and contribute to spacecraft systems design.

AI Research Intern — June 2026 - August 2026

Space Sciences Laboratory, UC Berkeley

- Developed computer vision and data analysis pipelines to analyze NASA Carruthers mission data, identify optical distortions, and validate spacecraft observations.

Stanford Small Group Scholars Program | Participant — October 2026 - Present

Standford University

- Selected for Stanford's Scholars Program, exploring data organization and labeling to improve AI predictions.

Software and IT Intern — June 2025 - August 2025

Marina Security Services, City: San Francisco

- Maintained and troubleshot IT equipment, workstations, and network connections. Maintained the company Website

Mathematics Tutor — August 2024 - Present

Berkeley City College

- Explain math concepts, work one-on-one or in groups with students to support them with mathematical questions.

Founder of Studentlife OS — Launched August 2025

- Designed and developed an AI-powered student success platform that centralizes internships, scholarships, research opportunities, and campus resources while leading product development and a team of interns.

Software and Hardware Intern — December 2024 - February 2025

Computer Repair- American Tech: City: Berkeley

- Repaired hardware issues and configured networks, routers, modems while troubleshooting connectivity problems.

LEADERSHIP, VOLUNTEER, AND AWARDS

Certificate of Hackathon Winner & Recipient of Winning Prize — Awarded January 2025

PepsiCo Scholarship & UNCF Scholarship & ASLC Scholarship recipient — Awarded April - August 2025

East Bay Regional Parks Volunteer Event Waiver - Volunteer — September 2024

Tutor - Cyber Defense - CyberSecurity - Cyberlaw Certificate — July 2024

6. References

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2. L'SPACE Program. "NASA L'SPACE: Lucy Student Pipeline Accelerator and Competency Enabler." https://lspace.asu.edu

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4. NASA. "NASA Revamps Challenge Linking Community College Studies to Aerospace Careers." September 2, 2026. https://www.nasa.gov/learning-resources/linking-community-college-studies-to-aerospace-careers/

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8. NASA Small Spacecraft Systems Virtual Institute. "State-of-the-Art of Small Spacecraft Technology: Guidance, Navigation, and Control." https://www.nasa.gov/smallsat-institute/sst-soa/guidance-navigation-and-control/

9. NASA. "Super Intelligence" (NASA artificial intelligence overview). https://www.nasa.gov/artificial-intelligence/

10. NASA Small Spacecraft Systems Virtual Institute. "State-of-the-Art of Small Spacecraft Technology: Power." https://www.nasa.gov/smallsat-institute/sst-soa/power-subsystems/

11. NASA Small Spacecraft Systems Virtual Institute. "State-of-the-Art of Small Spacecraft Technology: Small Spacecraft Avionics." https://www.nasa.gov/smallsat-institute/sst-soa/small-spacecraft-avionics/

12. NASA. "NASA Systems Engineering Handbook" (NASA/SP-2016-6105 Rev2). https://www.nasa.gov/reference/systems-engineering-handbook/

13. National Institute of Standards and Technology (NIST). "Cybersecurity Framework" (CSF 2.0). https://www.nist.gov/cyberframework

14. NASA Science. "New NASA Mission to Reveal Earth's Invisible 'Halo'." September 18, 2025. https://science.nasa.gov/science-research/heliophysics/new-nasa-mission-to-reveal-earths-invisible-halo

Logos shown belong to NCAS, L'SPACE, UC Berkeley Space Sciences Laboratory, and StudentLife OS. No NASA insignia are used in this portfolio.